

Module 3: Sewage Treatment

Sewage treatment is the process of removing contaminants from municipal wastewater, containing mainly household sewage and industrial wastewater. Physical, chemical, and biological processes are used to remove contaminants and produce treated wastewater (or treated effluent) that is safe enough for release into the environment. A by-product of sewage treatment is a semi-solid waste or slurry, called sewage sludge.

Decomposition of Sewage

(i) Aerobic Decomposition:

Aerobic decomposition is caused by aerobic bacteria. During this process, organic matter is broken up and oxidized to form stable end products such as carbon dioxide, nitrates, sulphates, etc. Sewage treatment units which work on aerobic decomposition process alone are aeration tanks, trickling filters, contact beds, oxidation ponds, etc.

(ii) Anaerobic Decomposition:

Anaerobic decomposition or putrefaction is caused by anaerobic bacteria. Anaerobic bacteria survive by extracting and consuming bounded molecular oxygen. The end products of anaerobic decomposition or putrefaction include gases like hydrogen sulphide, ammonia, methane, etc., and black residue. Sewage treatment units which work on putrefaction alone are septic tanks, Imhoff tanks, sludge digestion tanks, etc.

Cycles of Decomposition

1. Nitrogen Cycle:

The nitrogen cycle, which corresponds to the biochemical degradation or decomposition of nitrogenous organic matter, is an endless chain connecting the processes of life and decay of both animal and plant worlds.

(1) The nitrogenous organic matter, in the form of waste animal and plant matter, consisting of urea, proteins and hydrocarbons, undergoes decomposition either by way of oxidation by aerobic bacteria or by way of putrefaction by anaerobic bacteria, resulting in the formation of ammonia (NH_3) and other gases.

(2) By process of nitrification, ammonia is first converted into nitrites (NO_2) by partial oxidation and then finally to nitrates (NO_3) by the action of aerobic bacteria.

(3) The products formed in step (2) are consumed by plant life as food through photosynthesis. The plant life grows, due to which plant tissues, plant protein (seeds) and free oxygen are produced.

(4) The plant protein is consumed by animals, resulting in the production of animal proteins (meat, milk, etc.).

(5) The animal wastes in the form of urine and other excretions as well as dead bodies of the animals result in the formation of nitrogenous organic matter.

(6) The death or decay of plant life (as it happens in forests where leaves, flowers, etc. of plants lie in layers on ground surface to decay) may directly result in the formation of nitrogenous organic matter.

(7) Waste products such as urea and excretions of animals may sometimes decompose directly and form ammonia nitrogen.

(8) The nitrates formed in step (2) may be converted into or reduced to free nitrogen (and sometimes to ammonia) by anaerobic bacteria by a process known as de-nitrification.

(9) Free nitrogen produced in step (8) may directly be converted into plant proteins by certain type of bacteria present in the roots of plants, through a process known as nitrogen fixation.

2. Carbon Cycle:

The carbon cycle, which corresponds to the biochemical degradation or decomposition of carbonaceous organic matter, is also an endless chain connecting the processes of life and decay of both animal and plant worlds.

(1) The decomposition of carbonaceous organic matter, through oxidation by aerobic bacteria results in the release of carbon dioxide CO_2 , which is its final end product and is the stable form of carbon.

(2) Plants in the presence of sunlight and by the process of photosynthesis consume carbon from carbon dioxide produced in step (1). This results in the formation of plant carbohydrates, fats and proteins (sugars).

(3) The plants are consumed by animals, resulting in the formation of animal fats and proteins.

(4) Waste products as well as death of animals results in the formation of carbonaceous organic matter

(5) (a) Plant life gives off carbon dioxide at night through the process of respiration.

(b) Similarly animal life gives off carbon dioxide during respiration.

(6) Carbonaceous organic matter may also be formed directly by the death/decay of plant life.

3. Sulphur Cycle:

The sulphur cycle, which corresponds to the biochemical degradation or decomposition of sulphurous organic matter, is also an endless chain connecting the processes of life and decay of both animal and plant worlds.

(1) The decomposition of sulphurous organic matter, through the action of anaerobic bacteria, in the absence of oxygen, results in the formation of hydrogen sulphide (H_2S).

(2) By process of oxidation, hydrogen sulphide is first converted into sulphur and finally to sulphates.

(3) Sulphates, when consumed by plants, through photosynthesis, change into plant proteins.

(4) Animals consume the plant proteins which results in the formation of animal proteins.

(5) Wastes produced by animals and their dead bodies results in the formation of sulphurous organic matter.

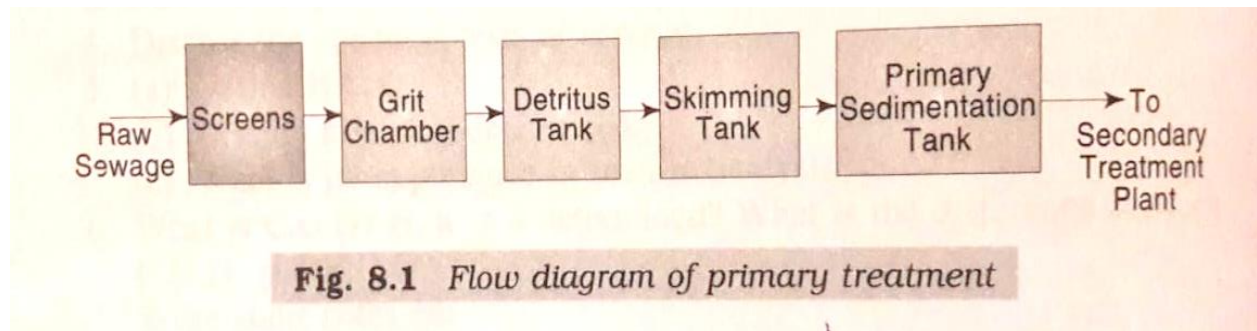
(6) Sulphates in the absence of oxygen, are converted into hydrogen sulphide (H_2S), by the process of reduction.

(7) Sulphurous organic matter may directly be produced by the death or decay of plants.

Sewage Treatment

1. Primary Treatment:

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Function of units: -

1. Screening: -

The screening is the first unit of primary treatment plant. The function of screen is to remove all the floatation debris like wood pieces, cloth and paper pieces, decayed fruits and vegetables, etc. if these floating matters are not eliminated, it may choke the pipe lines or it may cause damage to the pumping unit.

The screens may be constructed of M.S. bars or rods, gratings, wire meshes or perforated plates. The M.S. bar screen is made by placing parallel bars with spacings according to the following types:

(a) **Coarse Screen** :- The spacings of bars are more than 40 mm center to center.

(b) **Medium Screen** The spacing are less than 40mm.

(c) **Fine Screen** The spacings vary from 1.5 mm to 6 mm.

The screens may be fixed or movable. the inclination of the screen varies from 30° to 60° . The screens are placed at designed inclination in an oblong rectangular chamber. The ends of the chamber are tapered. It is constructed with brick masonry. The inner surfaces are plastered and finished with neat cement polish. A perforated rectangular channel is provided at the top of the screen for the collection of floating debris.

2. Grit Chamber: -

The function of grid chamber is to remove the inorganic substance like grit, sand, and other suspended materials. The velocity of flow (20-30 cm/sec) in the grid chamber is kept very low so that the detention period is available for the settlement of the above substances. add general design and dimension.

The grit chamber is an oblong rectangular chamber and constructed with brick masonry. the floor of the chamber is made sloping for the collection of grits at a particular zone. The inner surfaces are plastered and finished with neat cement polish. It consists of an agitator for agitating the deposited grits at the time of cleaning. A pipe line with valve is provided at the bottom of the chamber for periodical removal of the grits. The length, width and depth are designed according to the volume of sewage.

3. Detritus Chamber:-

The function of detritus tank is to remove finer organic and inorganic particles. Practically, the detritus tank is similar to grit chamber. The only difference is that the grid chamber is meant for removing the large particles and detritus tank is meant for removing finer particles.

It is a rectangular tank with vertical sides and tapered bottom and constructed with brick masonry. The depth of the tank varies from 2-3 m and the length varies from 20-40 m. The width depends on the volume of sewage to be treated. The velocity of flow varies from 15-20 cm per second and the detention period varies from 2 to 3 minutes.

4. Skimming Tank:-

Normally, the grease and oil are arrested by the grease and oil trap before the sewage enters the sewer line. But, these elements cannot be completely eliminated by trap. The skimming tank is provided to remove the remaining portion of grease and oil from the sewage. Otherwise, these oily substances will cause disturbances in activated sludge process and tricking filter.

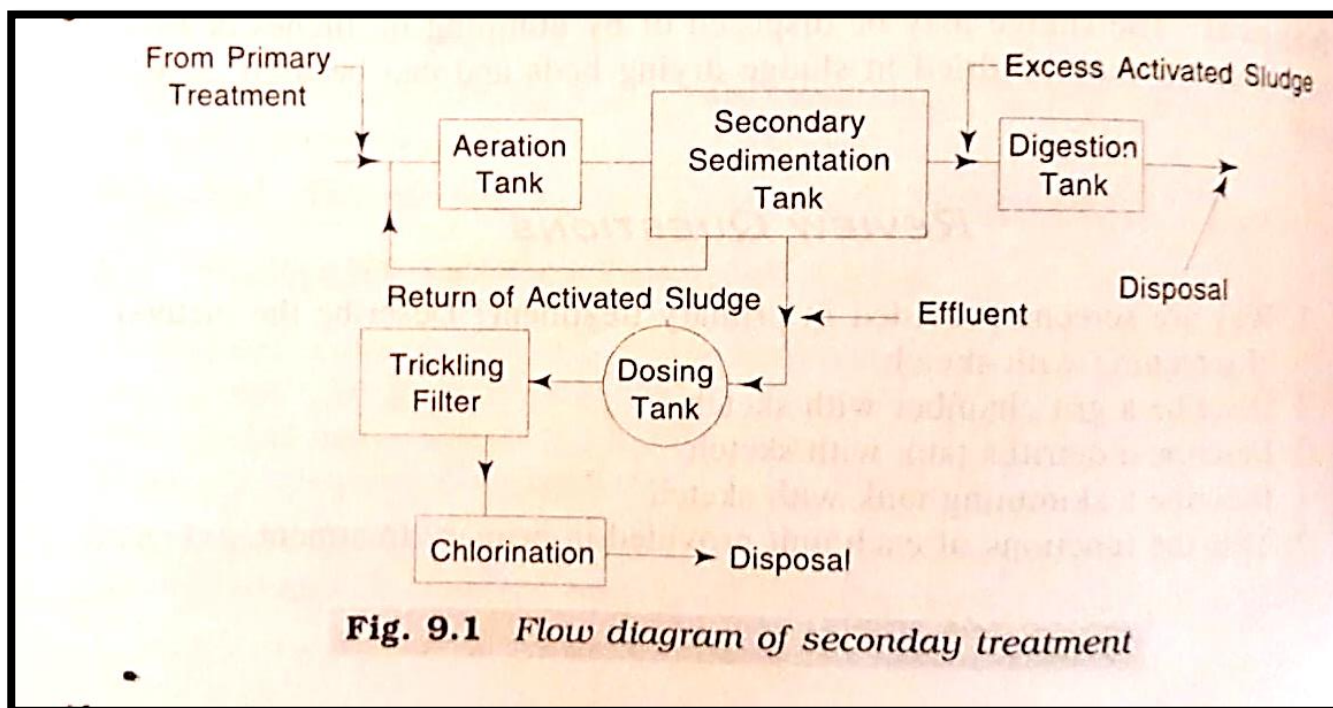
It is a rectangular tank constructed with brick masonry. Air diffusers are provided at the bottom of the tank through which compressed air is sent to form bubbles in the sewage. Bent pipes are provided at the inlet and outlet sides. C.I. covers are provided at the top for the removal of scum of grease and oil.

5. Primary Sedimentation Tank:-

The primary sedimentation tank is also known as primary clarifier or primary settling tank. The purpose of this unit is to remove colloidal particles like silt and clay and some organic substance. Moreover, it reduces the load on the secondary treatment. coagulants may be used, if necessary.

It is a rectangular tank constructed with brick masonry. Baffle walls are provided in zig zag way so as to lengthen the path of the flow of sewage. Inlet and outlet pipes are provided on opposite corners and these are provided with valves. A sludge removal pipe is provided at the bottom of the tank.

2. Secondary Treatment:



Functions of the Units of Secondary Treatment

1. Aeration Tank

the process of absorbing oxygen from air is known as aeration. The aeration of sewage is the most vital operation in activated sludge process. Due to aeration, the sludge containing bacteria becomes more powerful in decomposing the sewage quickly. Hence, the aeration is the first step of secondary treatment.

The aeration is achieved by providing air diffusers in aeration tank. The air diffusers may be of the following three types:

- (a) Jet Diffuser
(b) Plate diffuser
(c) Tube diffuser (60cm long and 7.5 cm dia)

Type of aeration tank

- (a) Rigid and furrow type tank (10 m long x 5m wide x 3 m deep)
(b) Spiral flow type tank

2. Secondary Sedimentation Tank

The secondary sedimentation tank is also known as secondary clarifier or Secondary settling tank.

It is a rectangular tank constructed with brick masonry. A baffle wall is provided in the tank to divide it into two zones. The floor of each zone is made sloping towards the side. The inlet and outlet arrangements are made through the chambers and the sludge removal pipes are provided on both sides. Generally, the length of the tank is three times the width. The detention period is generally 1 hour.

The aerated sewage is allowed to enter the tank through the inlet chamber and it is detained for one hour in the tank. The sludge is deposited at the bottom of the tank. This sludge becomes activated due to aeration. The effluent is discharged through the outlet chamber and taken to the trickling filter.

3. Sludge Digestion Tank

The decomposition of complex organic matters in sludge by the bio-chemical reactions created by anaerobic bacteria is termed as sludge digestion.

Necessity of Digestion

The following are the necessity of sludge digestion:

- (i) To destroy pathogenic bacteria.
- (ii) To reduce the volume of sludge so that it can be disposed of easily.
- (ii) To obtain combustible gases.
- (iv) To obtain good fertiliser.
- (v) To reduce the moisture content for the facility of handling and transporting.

The sludge can be digested by the following methods:

(a) Sludge digestion tank

The enclosure tank is generally circular in shape and is constructed with R.C.C.

The diameter of tank varies from 5-20 m and depth varies from 3-5 m. However, the actual size depends on the volume of sludge to be treated. The floor of the tank is made sloping like hopper and the slope is generally 1: 2 or 1: 3.

The following are the working principles of sludge digestion tank:

- (a) The raw sludge is allowed to enter the tank through the inlet pipe and is thrown at the centre of the tank.
- (b) The sludge is digested by the decomposition of complex organic matters by anaerobic bacteria.
- (c) The digested sludge is settled at the bottom of the tank which is withdrawn through the outlet valve and left for drying. The gases (mainly methane) are collected at the dome. The gases are withdrawn through the outlet pipe and are used as fuel.
- (d) The supernatant liquid is collected at the space between the digested sludge zone and the gas dome. This liquid is withdrawn from different levels and disposed of in the natural water course.

(b) Imhoff tank

In Imhoff tank, the sedimentation and digestion are carried out simultaneously.

The following are the parts and working of Imhoff tank. Shape It is circular in shape with hopper like bottom and constructed in R.C.C.

- (a) Sedimentation chamber
- (b) Digestion Chamber

- (c) Gas chamber
- (d) Neutral Chamber

1. Dosing Tank

The sewage is not applied on the trickling filter continuously. It is applied a by automatic siphonic device provided with the dosing tank. It consists of a U-pipe with its vertical end inserted into the tank. A hood is provided over the free end of the pipe. The other end of U-pipe is connected to the inlet pipe of trickling filter. When the level of sewage remains on the line b - b, there will be no siphonic action. When the level rises and reaches the line a - a, the siphonic action starts. The siphonic action continues until the level reaches to b - b. The rate of flow is adjusted in such a way that the dosing of the filter is achieved at every 10 minutes.

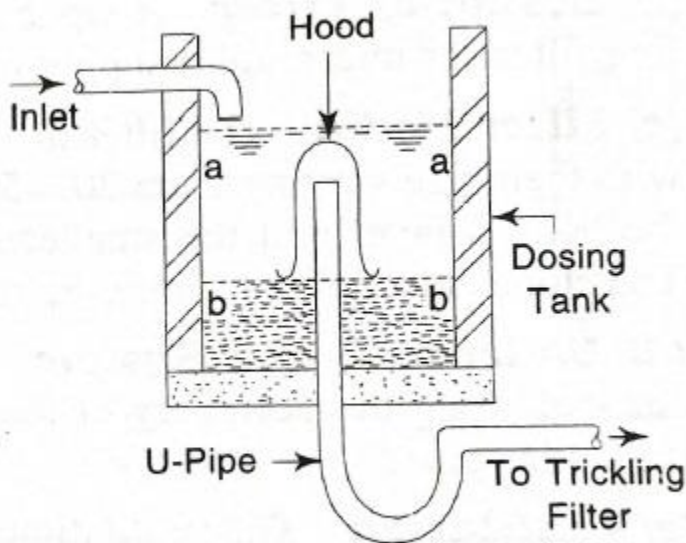


Fig. 9.9 Dosing tank

2. Trickling Filter

The theory of trickling filter is based on the principle that the bacterial film (known as bio-film) which is formed around the filtering media, is the source of formation of the bacterial colonies. These bacteria decompose the organic matters for their survival. So, the trickling filter serves the purpose of breaking the complex organic matter by fertilising the bacteria.

Generally, the trickling filter is circular in shape. It consists of four numbers of rotary distributing arms which have perforations at the bottom. The arms are fitted with a central support which is rotated by a suitable device. The floor of the filter is made of concrete and its slope is made towards the periphery.

3. Chlorination of Sewage

Necessity

The chlorination of sewage is done for the following reasons:

- (a) It prevents the growth of algae and other plant life in sewers.
- (b) It prevents the corrosion of sewers.
- (c) It protects the aquatic life.
- (d) It eliminates the odour from effluent.

Chlorination Tank

The chlorination tank is a rectangular chamber constructed of brick masonry. It consists of a vertical shaft and horizontal shaft. The horizontal shaft consists of several paddles which are rotated by a driving unit. The dosing of chlorine is done in a dosing chamber through a funnel. Generally the dose varies from 10 to 20 ppm and the contact period is 15 minutes.

Method of sewage disposal

The sewage can be disposed of by two ways:

- .(a) By artificial method
- (a) By natural method.

In artificial method, the sewage is conveyed through the underground sewer line and taken to the treatment plant where it undergoes primary treatment and secondary treatment. Then the effluent is discharged into the natural water course.

In natural method, the sewage is conveyed through the underground sewer line and it is simply disposed of to the natural water course or some other way. The following are the natural methods of sewage disposal:

(a) Dilution method -

The term dilution is meant by the fact that the disposal of sewage is done by discharging it into the natural water courses such as river, stream, lake or sea. By dilution, the sewage is purified by the self-purification capacity of the natural water. Again, success of self-purification depends on the dilution factor and the nature of sewage at the time of disposal-whether it is fresh or stale. The dilution factor is defined as the ratio of the amount of diluting water to that of the sewage. The dilution factor should be ascertained by measuring the discharge of the river throughout the year. At the dilution factor above 500, the sewage requires no treatment. It can be discharged directly into the river. If the dilution factor becomes less than 500, primary treatments should be given to the sewage before discharging.

The following points should be remembered while adopting the dilution method:

- i) The city or town should be situated near the sea or large river.
- ii) The river should be perennial and plenty of water should be available throughout the year.

- (iii) The current of the river should be considerable so that the sewage is carried quickly towards the downstream.
- (iv) The sewage should be fresh at the time of disposal or it should be discharged within 3 hours of its formation.
- (v) During the period of flood or heavy rainfall, the natural water course should not exert back pressure towards the sewer.
- (vi) Laboratory test should be carried out with the sample of water at the down stream side of the point of disposal to verify the quality-whether the water is fit for bathing, washing, irrigation, etc. or not.
- (vii) If the sewage contains industrial wastes, then it should be verified to ascertain the quality of water so that it may not be detrimental aquatic life.

(b) Sewage farming method-

When sewage is applied on agricultural land for the growth of crops, then it is termed as sewage farming. The sewage contains much fertilizing elements such as nitrates, sulphates and phosphates. These elements are extracted from the soil by the roots of the plants.

Conditions of Sewage Forming

The following conditions should be remembered while providing the method of sewage farming:

- (i) The farm should be located far away from the locality, because it may create bad smell and insanitary condition.
- (ii) The hygienic safety of the workers should always be observed to protect them from the bad effect of pathogenic bacteria.
- (iii) The raw sewage should never be supplied to the farm.
- (iv) It is better to apply the sewage after primary treatment.
- (v) The working of the farm should run on constant supervision, so that insanitary condition may not arise due to over irrigation.
- (vi) Precautions should be taken to avoid sewage sickness.

Application of Sewage

The sewage may be applied on the land by the following methods:

- (a) Surface irrigation system
- (b) Sub-surface irrigation system
- (c) Sprinkler irrigation system

Sludge disposal

The sludge (i.e. the settleable part of the sewage) has an objectionable odour and it possesses the property of pollution if not properly disposed of. But, it is dried and properly stored, then it can be used as a good manure. The sludge may be of various types, such as:

- (a) Sludge from primary settling tank
- (6) Sludge from secondary settling tank
- (c) Sludge from trickling filter
- (d) Sludge from digestion tank

The sludge obtained from all the sources may not possess same characteristics. So, the unwanted sludge should be disposed of. The following are the various methods of sludge-disposal:

(a) Disposal by drying bed:

It is a rectangular pit excavated to the required depth. It is divided into several beds. The beds are filled with gravel and coarse sand as filter. The depth of gravel is 30 cm. The bottom layer of 15 cm consists of 40 mm gravel and the top layer consists of 6 mm gravel. The depth of coarse sand is 15 cm. The underdrainage system consists of a pipe line perforated at the top,

The working principle of drying bed is stated below:

- (a) The sludge is spread evenly over the sand bed to a thickness of 30 cm.
- (b) After 7 days, the surface of sludge cracks which indicates that the sludge is ready for removal.
- (c) The sludge cakes are removed by a spade and stored suitably for the use as manure.
- (d) This method is generally recommended for the black sludge (i.e. humus) which has good manurial value.

(6) Disposal on land

The sludge may be disposed of to the barren lands to make it fertile in due course. The sludge may be applied in barren lands in the following ways:

- (a) **By Ploughing** The sludge is spread over the land and ploughed by tractor. It is then left for several months. Again, the land is ploughed by Spreading a fresh layer of sludge. Thus the barren land may be converted to fertile land.
- (b) **By Trenching** Trenches may be excavated in parallel rows at an interval of 1m on the barren land. The trenches are filled up with sludge and a thin layer of excavated earth is spread over it. After one month, new trenches are excavated perpendicular to the previous trenches and these are filled up with sludge as previous method. Thus the barren land is converted to fertile land where plenty of crops may be grown.

(c) Disposal by throwing into sea

In coastal towns or cities, the sludge may be directly thrown into the sea. Because, there is no utility of sludge, not even as manure or for land treatment. It is a very easy and cheap method of sludge disposal. While dumping into the sea, the following points should be remembered:

- (a) The sludge should be thrown into the deep sea far away from the shore.
- (b) The direction of wind and wave should be carefully studied so that the sludge may not return to the shore.
- (c) The sludge should not be thrown in the zone where the fishermen catch sea-fishes.

(d) Disposal by lagooning

A lagoon is an artificial pond of shallow depth of about 1 m. The surface area of lagoon depends on the volume of sludge and the availability of land. Embankments are constructed with the excavated earth. A layer of sand of thickness 15 cm is laid at the bottom of the trench. The lagoon is filled up with wet sludge and left for few months. The sludge is dried gradually and cracks are formed on the surface. Then the dried sludge is removed and used as manure. Thus is a very cheap method, if sufficient land is available.

(e) Disposal by incineration.

The term incinerator indicates the process of burning to ashes. The sludge is burnt in incinerator which may be of the following types:

(a) Flash Type

This type of incinerator consists of a tower. The tower is first heated by fuel (generally coal) and then the wet sludge is thrown from the top of the tower. The hot gases and the moisture are removed through the top of tower. The burnt sludge is collected at the bottom. It is then disposed of by dumping in ditches or low-lying areas.

(b) Multiple Hearth Type

This type of incinerator consists of a furnace. The furnace consists of a series of hearths placed in successive layers. The wet sludge is placed over the top hearth. Then the sludge rolls over the successive hearths towards the bottom hearth. The gases and moisture are removed and the burnt sludge is collected at the bottom. It is then disposed of by dumping in ditches or low-lying areas.

3. Tertiary Treatment:

Tertiary treatment aims at further purification of waste water and also for its recycling. The number of methods used for tertiary treatment is:

- (i) Chemical coagulation and filtration
- (ii) Adsorption
- (iii) Chemical oxidation.

- (iv) Desalination,

Desalination, or desalting, is the separation of fresh water from salt water or brackish water. Major advances in desalination technology have taken place since the 1950s, as the need for supplies of fresh water has grown in arid and densely populated areas of the world. Desalted water is the main source of municipal supply in areas of the Caribbean, the Middle East, and North Africa, and its use is increasing in the southeastern United States. Although it is relatively expensive to produce, desalted water can be more economical than the alternative of transporting large quantities of fresh water over long distances.

There are two basic types of desalting techniques:

1. thermal processes

2. membrane processes.

Both types consume considerable amounts of energy. Thermal methods involve heat transfer and a phase change of the water from liquid into vapour or ice. Membrane methods use very thin sheets of special plastic that act as selective barriers, allowing pure water to be separated from the salt.

(v) Oxidation ponds.

The oxidation pond is an excavation of rectangular ditch of shallow depth. The sewage is stored in this pond for a considerable time. During this period, the sewage is decomposed by the action of aerobic bacteria, algae and sunshine. That means, it is a natural method of sewage treatment. The aerobic bacteria absorb oxygen from the atmosphere for their survival and break up the organic matters in sewage to simple stable compounds.

The recycled waste water can be used for irrigation, mainly for non-food crops such as grasslands, lawns, play grounds, etc.